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- (b) waves travelling in opposite directions overlap / incident and reflected waves overlap
 (allow superpose or interfere for overlap here) B1
 waves have the same speed and frequency B1 [2]

- (c) (i) time period = 4×0.1 (ms) C1
 $f = 1 / T = 1 / 4 \times 10^{-4} = 2500$ Hz A1 [2]

- (ii) 1. the microphone is at an antinode and goes to a node and then an antinode / maximum amplitude at antinode and minimum amplitude at node B1 [1]
 2. $\lambda / 2 = 6.7$ (cm) C1
 $v = f\lambda$ C1
 $v = 2500 \times 13.4 \times 10^{-2} = 335 \text{ m s}^{-1}$ A1 [3]

incorrect λ then can only score second mark

- 7 (a) (i) the half life / count rate / rate of decay / activity is the same no matter what